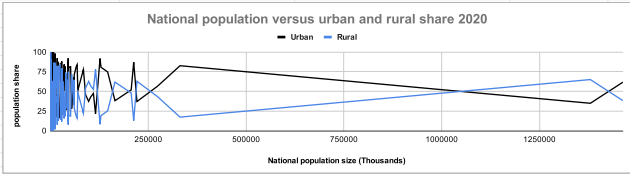


Summarise the national population size to better understand how the dataset represents the entire world population.					
National Population Size 7,786,695,107.75	World Population (estimated) 7,821,000,000.00	Population of Urban Areas 4,375,308,462.55	Population of Urban Areas (estimated) 4,301,550,000.00	Urban Share (%) 56.19%	Urban Share (% estimate) 55.00%
percentage difference					
Percentage diff (pop_n) 0.44	Percentage diff (urban share) 2.14	Percentage diff (urban value) 1.70			

visualisation to compare the share of the national population living in urban versus rural areas.



We find that our data visualisation is not very comprehensible, due to some of our pop_n values being much larger than other values, in other words, outliers.

To deal with the outliers in our dataset and thereby increase the readability of our data visualisation, we change the unit of our x-axis.

Advantage: Our visualisation is much more comprehensible and we don't lose information in our dataset for any future analysis we might need to do.

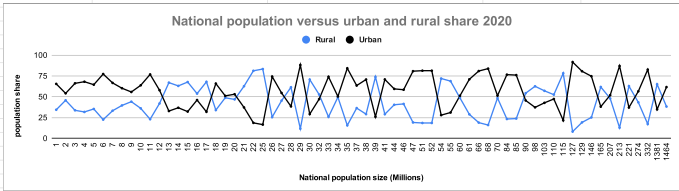
Disadvantage: Our x-axis doesn't represent the true difference between population sizes.

Another solution is not recommended

Delete the outliers from our dataset. Advantage: We don't have to change anything in our data visualisation as it would update automatically. Disadvantage: We lose information about some of the countries included in the dataset in this visualisation and any future analysis we might need to do.

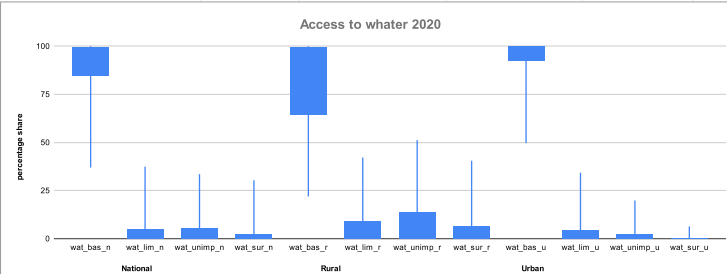
We edit our visualisation by adding a maximum cut-off value to our x-axis.

Advantage: Our visualisation is more comprehensible and we don't lose information in our dataset for any future analysis we might need to do. Disadvantage: Our visualisation doesn't represent the entire dataset, only a subset of it.

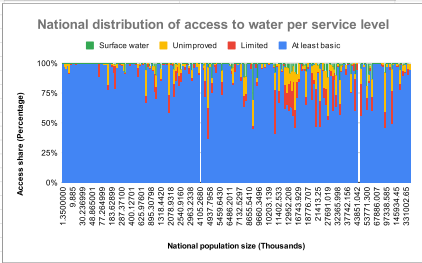


	Max	Min	mean	median	mode	q1	q3	IQR	std
wat_bas_n	100	37	11.65400597	97	89	85	99	14	15.10911192
wat_lim_n	37.42696287	0	4.823871686	0.4707829181	0	0	4.851219157	4.851219157	6.969846633
wat_unimp_n	33.53911377	0	5.20197667	0.8536474704	0	0.03390622265	5.279309022	5.245403699	7.174477036
wat_sur_n	30.36979308	0	2.593862098	0	0	0	1.878301993	1.878301993	3.923973987
wat_bas_r	100	21.98279234	18.25095781	90.73161683	100	64.8254211	98.11924249	34.29382139	21.50663741
wat_lim_r	42.16438068	0	6.460732214	1.812330835	0	0	8.553175126	8.553175126	8.704765846
wat_unimp_r	51.21598167	0	9.226431011	3.229749713	0	0.1584109169	13.2663393	13.10792838	11.82849434
wat_sur_r	40.51813242	0	4.982307148	0.2196897375	0	0	6.157471481	6.157471481	6.847951395
wat_bas_u	100	49.66166495	5.920278573	98.10662849	100	92.56287752	99.94941803	7.38654051	8.03886161
wat_lim_u	34.27978009	0	3.958164273	0.5	0	0	3.966347184	3.966347184	5.63506705
wat_unimp_u	19.83580681	0	2.054124356	0.3498740029	0	0	2.059001302	2.059001302	3.23457032
wat_sur_u	6.362160532	0	0.4640144272	0	0	0	0.1597056849	0.1597056849	0.8696316426

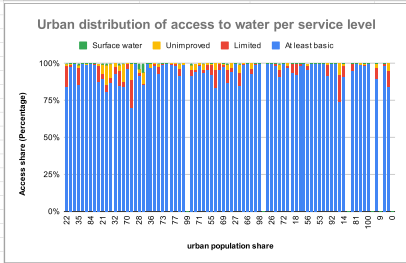
For 'wat_bas_r_u', we see that the mean is relatively high and the interquartile range is about a tenth of the range, which indicates that our data are concentrated around a point closer to 100% than 0%. This means that the majority of people represented in the data have access to at least basic water services on a national level.



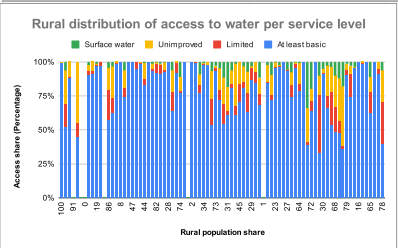
Visualise the national access to water on all four levels based on the national population size



Visualise the urban access to water on all four levels based on the urban population.



Visualise the rural access to water on all four levels based on the rural population.



income_group	sum of pop_n	avg of pop_u	avg of wat_bas_n	avg of wat_lim_n	avg of wat_unimp_n	avg of wat_sur_n
High income	1212076.066	79.35545728	69.56019792	0.1769010855	0.2408458913	0.02205537705
Low income	590425.1621	36.04465336	62.91937356	16.54838621	15.21093083	5.421307393
Lower middle income	3395310.231	48.78990351	82.21042088	5.688425074	7.898143786	4.28534813
NAN	37264.34029	61.46049964	97.1814662	0.1482198143	2.386919485	0.3238794261
Upper middle income	2547619.307	64.68569565	96.430472	1.561189554	1.483406065	0.572653552
Grand Total	7786695.108	61.88772329	89.86478253	3.865265345	4.415010654	1.915488584

